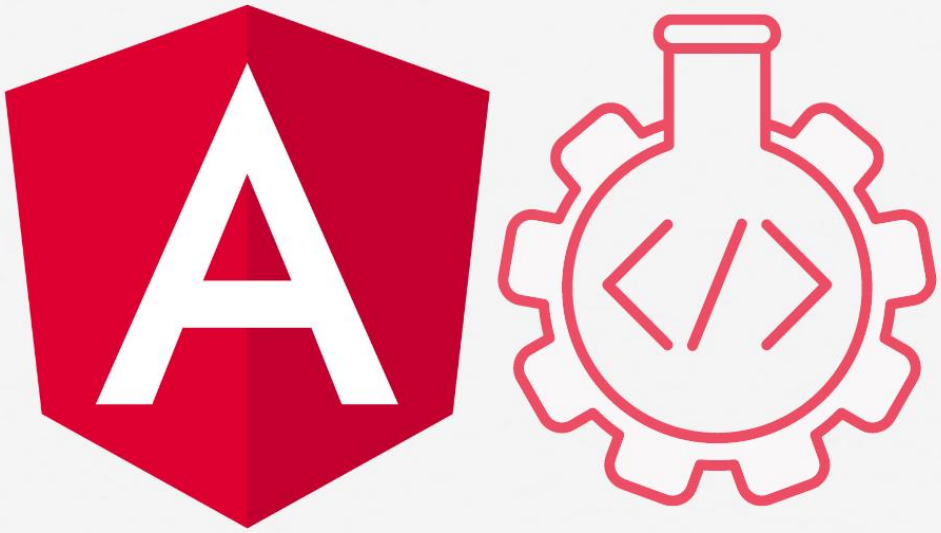
Angular Certification Training Lab Guide



ScholarHat

Angular Training: Coding Lab Guide

Our learn-by-doing method enables you to build practical/coding experience that sticks. 95% of our learners say they remember more when they learn by doing hands-on which is required to translate knowledge into real-world results.

The recommended hardware and software requirements for our training are given below:

Hardware Requirements

- 1.6 GHz or faster processor
- 4 GB or higher RAM
- 20 GB of available hard disk space
- DirectX 9-capable video card that runs at 1024 x 768 or higher display resolution

Software Requirements

- Windows 10 or higher, Mac or Linux,
- Visual Studio Code Latest Version
- Node.js 16.x or higher
- Git
- Install Angular CLI after installing Node.js by running the command:
    - npm i -g @angular/cli
- Latest Browser i.e. Google Chrome/ FF/IE
- Internet Connectivity

Software Download Links

- Visual Studio Code: http://code.visualstudio.com
- Node.js: https://nodejs.org/en
- Git (for Windows): https://git-scm.com/download/win

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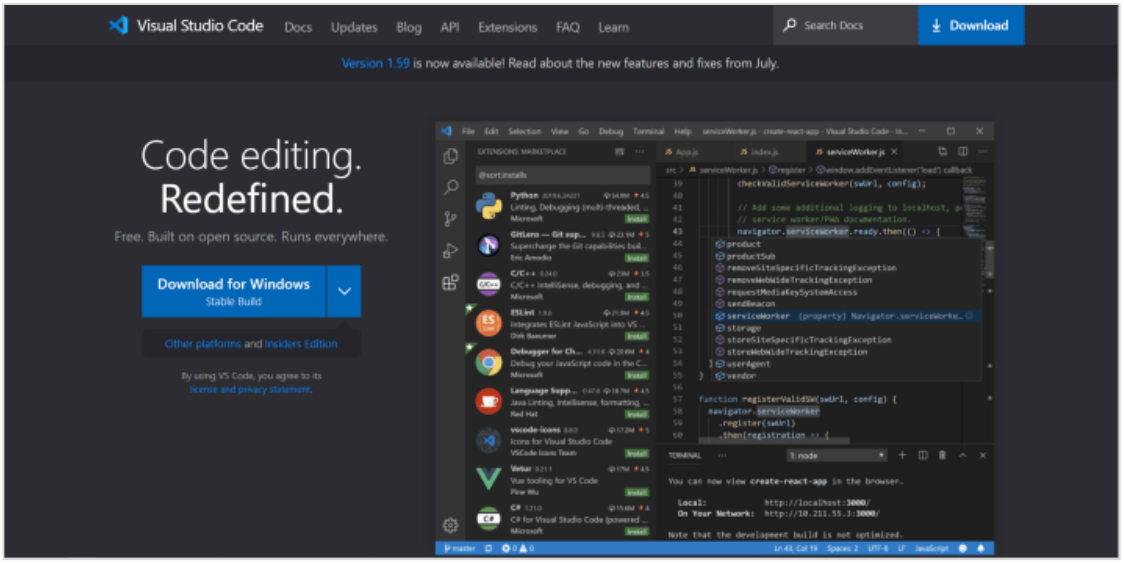
1. Visual Studio Code Installation

VS Code is an open-source code editor developed by Microsoft for Windows, Linux, and Mac OS platforms. Here we discuss the steps to install Visual Studio Code or VS code on Windows. It has built-in support for TypeScript, JavaScript, Node.js, and almost all programming language/library/framework-based plugins are available in VS studio right now.

- **Step 1: Go to https://code.visualstudio.com/ and download the Setup file according to your Operating System.**



- **Step 2: After Downloading the file from the link, open the installer.**
- **Step 3: Accept the license agreement and click on Next.**

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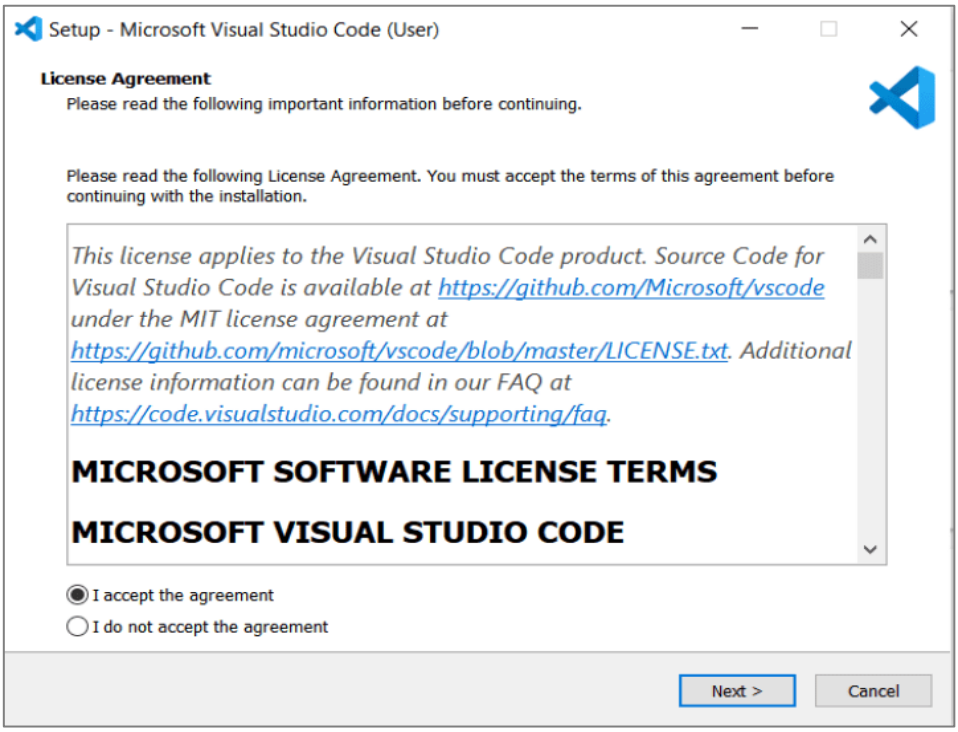
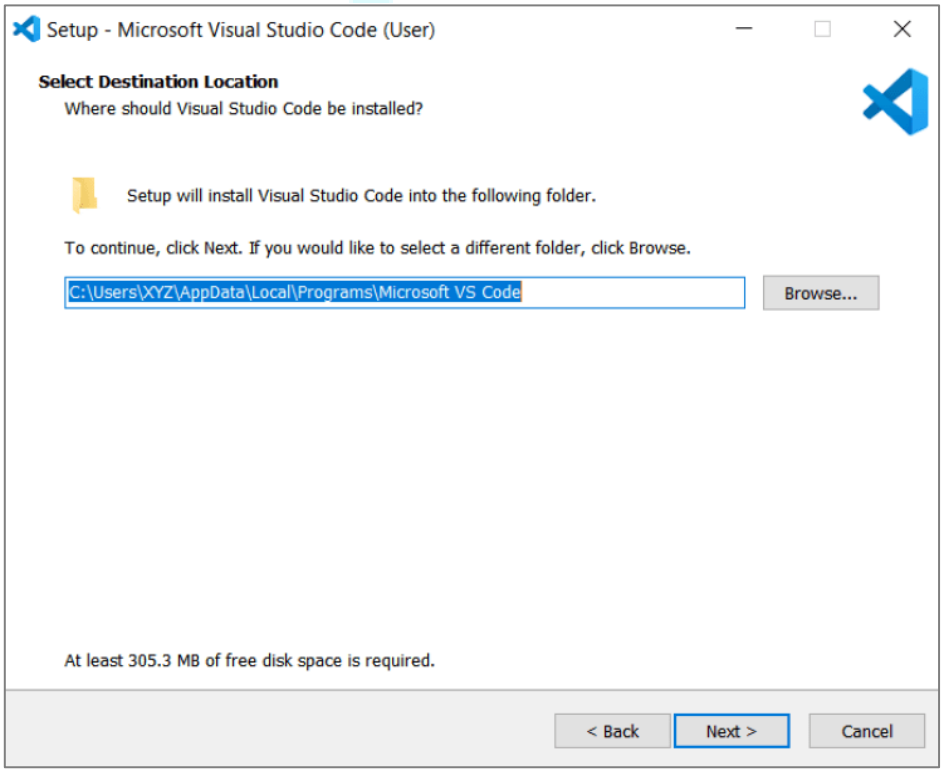


- **Step 4: You can Select the destination where you need to install, by default it will be installed in the folder given in the below image.**



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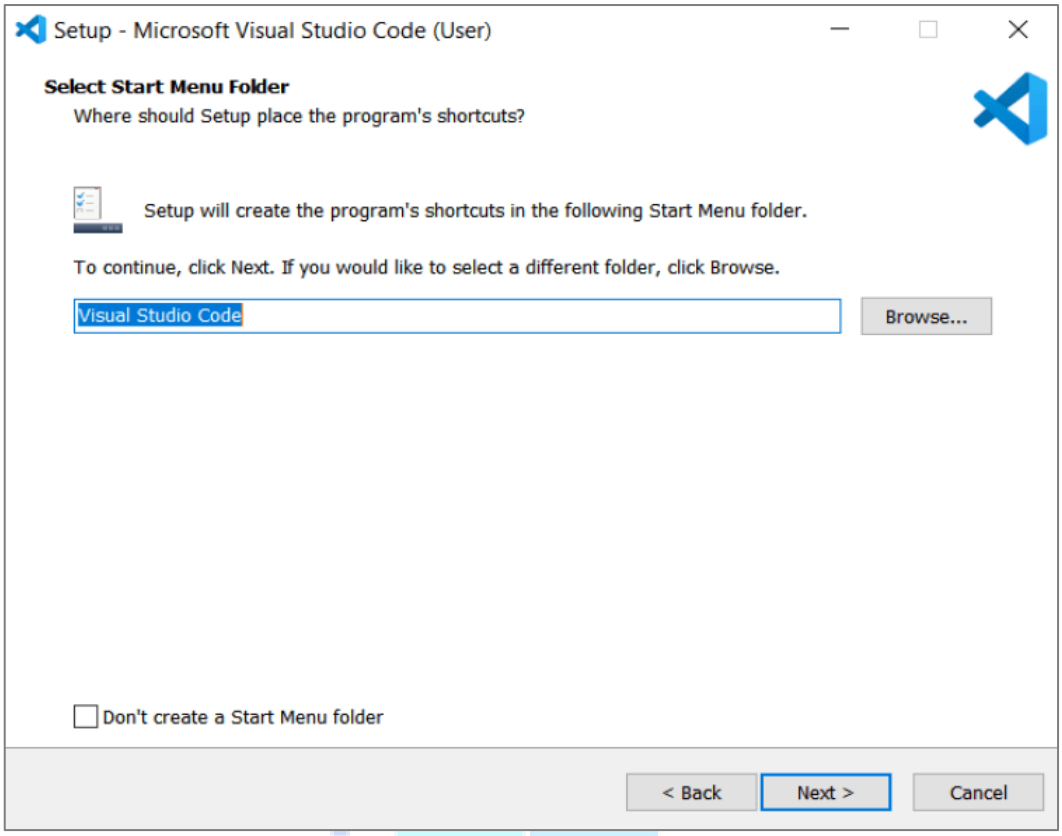
- **Step 5: Click on Next and It will create folder in start menu.**



Step 6: Click all the checkboxes and it will be helpful and it will also add to the PATH.

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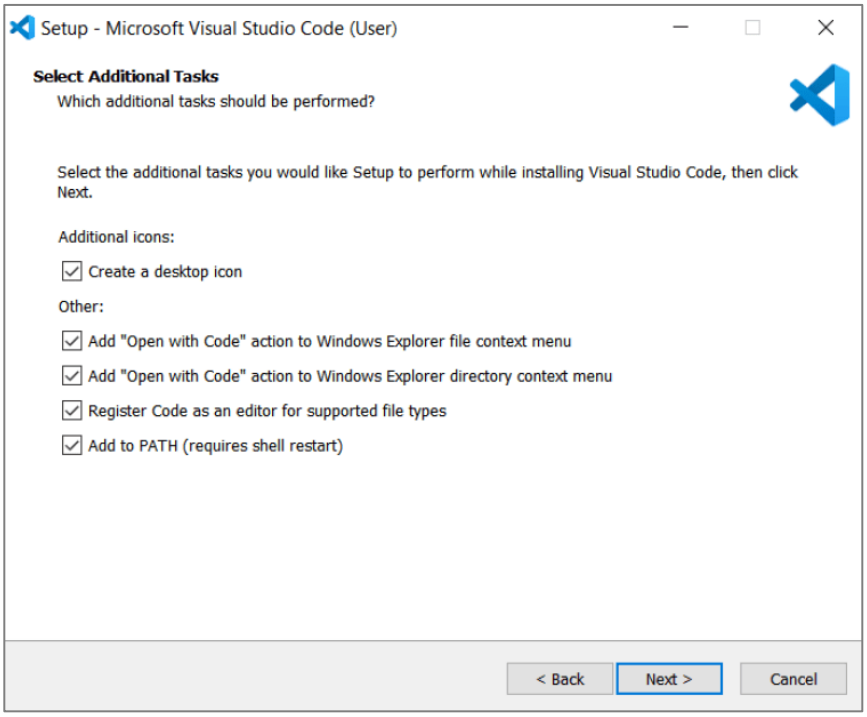
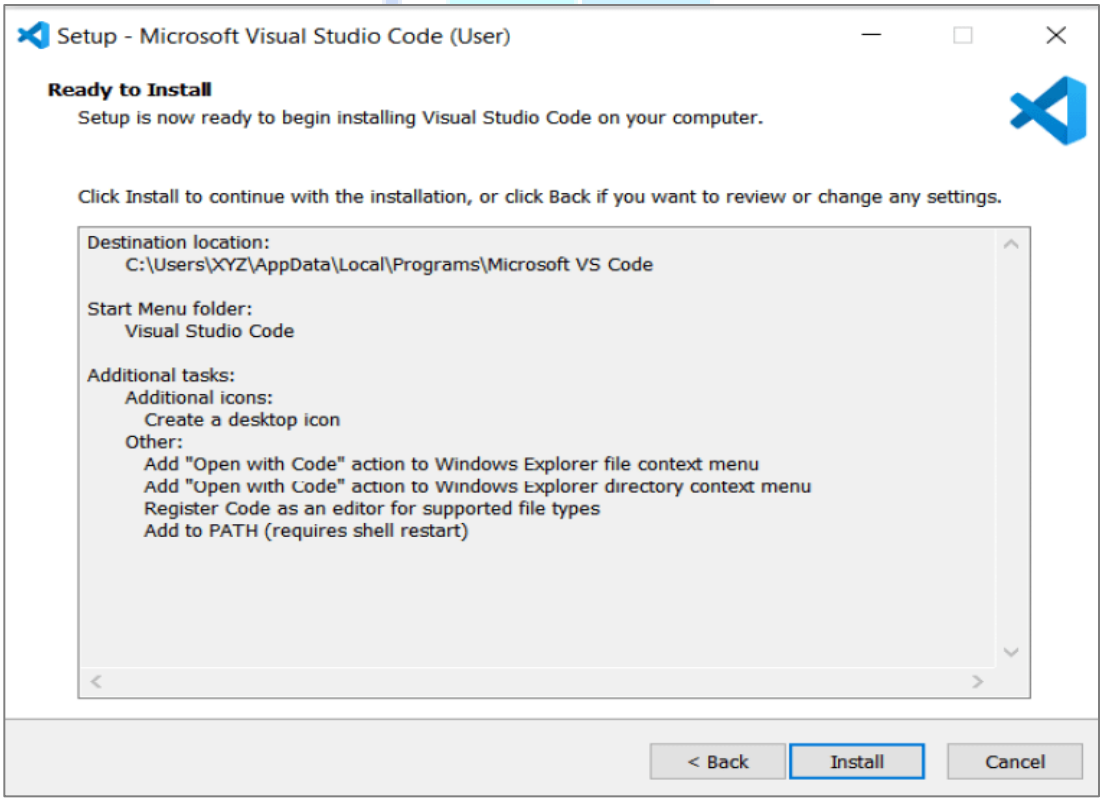


- **Step 7: Review Settings**



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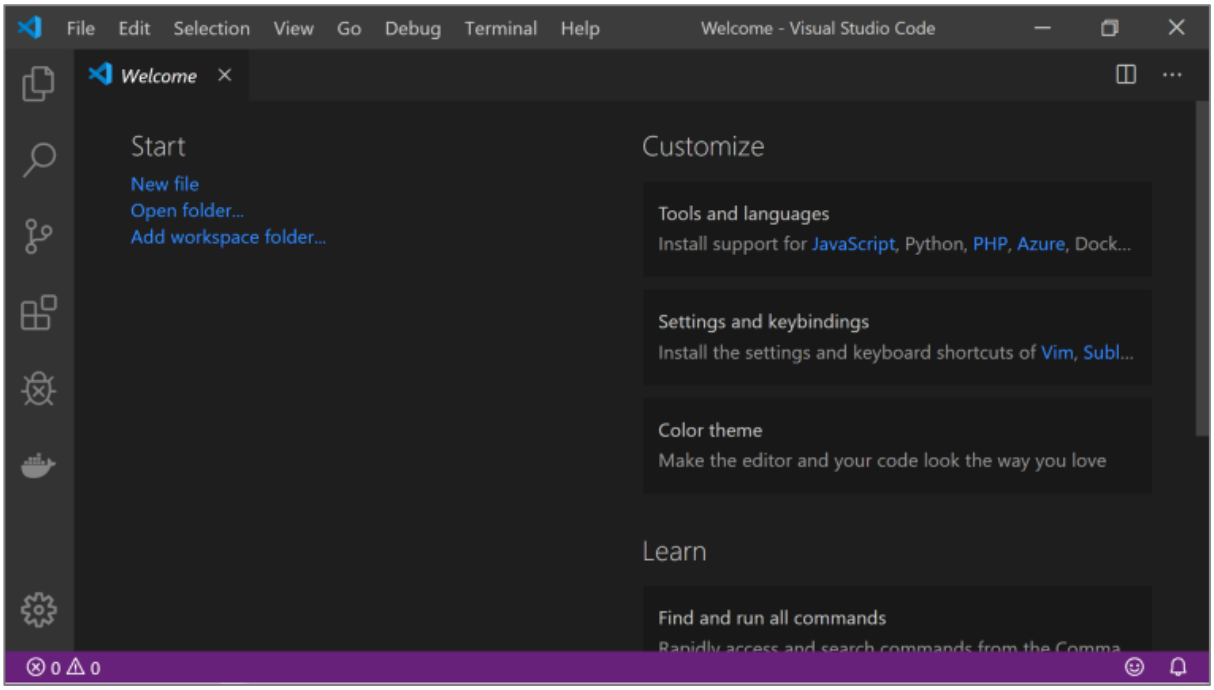
Now it will show all your settings there to install VS Code.

- **Step 8: Launch VS Code**

Then you will see the Welcome Screen of VS Code as shown below.



Visual Studio Code has successfully installed.

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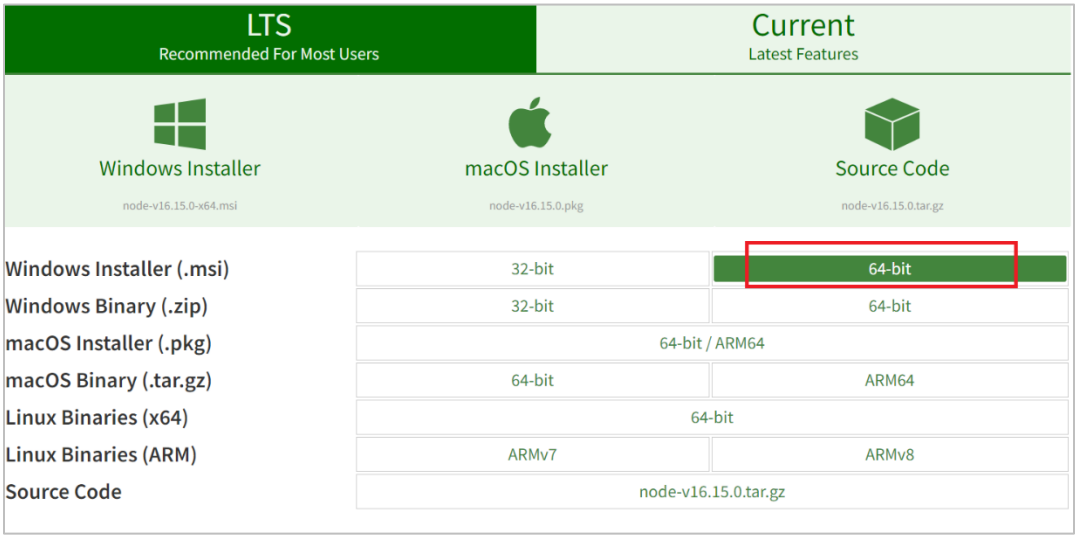
2. Node.js Installation

Node.js is a JavaScript runtime environment that allows you to execute code outside of a browser. node.js also provides a library of modules that makes it easy to create web applications.

- **Step 1: Install Node.js**

Follow the link: https://nodejs.org/en/download/ Download the Node.js installer for Windows and install it.



- **Step 2: Open Setup File**

Now from Setup file, Start the Installation Wizard.

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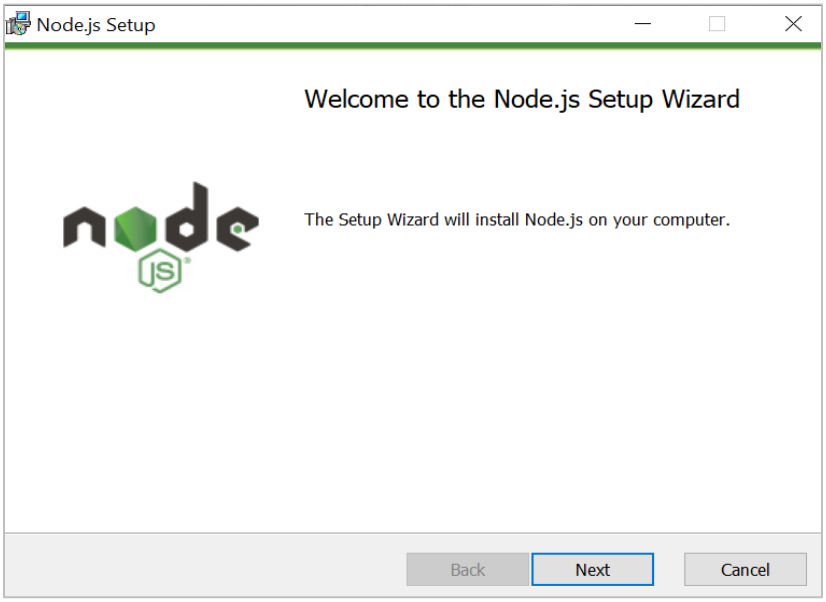
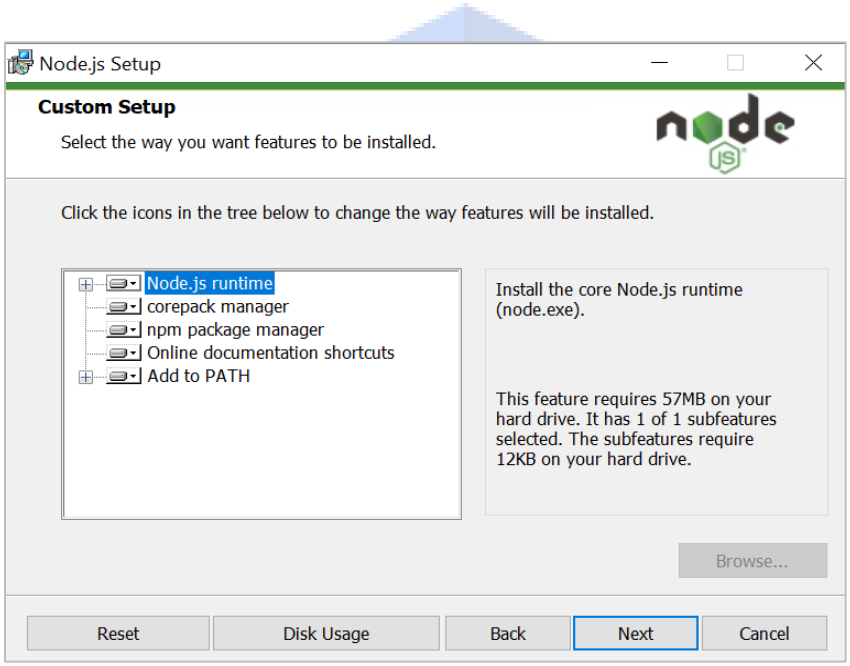


- **Step 3: Click Agree and Make sure all the options are installed.**



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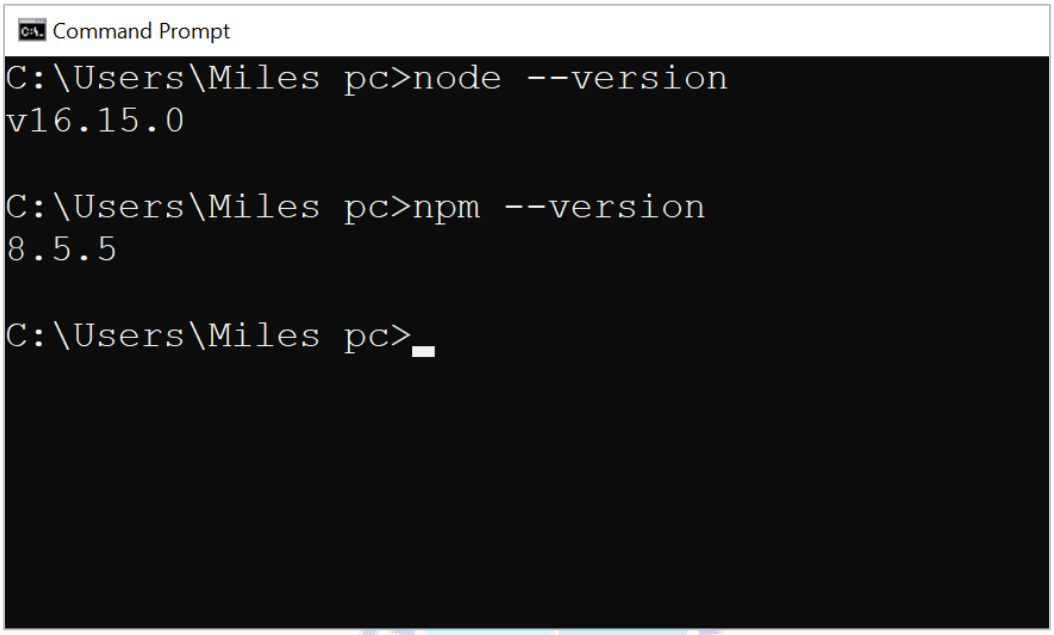
- **Step 4: Finish the installation and open Command prompt/Terminal.**

To verify the installation, Type node –-version and npm –-version to check the versions of Node and NPM.



3. TypeScript and Angular Installation

Angular is a client-side web framework created by Google. It allows developers to create single-page applications that are responsive and fast. Angular is built on TypeScript, and it uses a component-based architecture. This means that applications are divided into small, self-contained pieces that can be reused in other angular applications.

- **Step 1: Install TypeScript.**

Open the link https://www.npmjs.com/package/typescript

Copy the above command *"npm install -g typescript"* and run it on the command prompt.

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```
Command Prompt
Microsoft Windows [Version 10.0.19044.1645]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Miles pc>npm install -g typescript

changed 1 package, and audited 2 packages in 2s

found 0 vulnerabilities

C:\Users\Miles pc>
```

- **Step 2: Install Angular CLI (Command Line Interface).**

Run the following command to install Angular CLI.

npm install -g @angular/cli

```
Command Prompt
C:\Users\Miles pc>npm install -g @angular/cli

added 196 packages, and audited 197 packages in 26s

24 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Miles pc>
```

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- **Step 3: To check if your Angular CLI is successfully installed, Run “ng --version” on your Command Prompt.**

```
C:\WINDOWS\system32\cmd.exe
C:\Users\Miles pc>ng --version

Angular CLI: 13.3.4
Node: 16.15.0
Package Manager: npm 8.5.5
OS: win32 x64

Angular:
...

Package                      Version
---------------------------------------------------------
@angular-devkit/architect    0.1303.4 (cli-only)
@angular-devkit/core         13.3.4 (cli-only)
@angular-devkit/schematics   13.3.4 (cli-only)
@schematics/angular          13.3.4 (cli-only)
```

Your Angular CLI has successfully installed.

- **Step 4: Type “ng new angular-fundamentals” and hit enter to create the Angular Fundamentals app.**

After hitting enter, you will be asked to answer two questions.

Answer the questions as shown in the image below.

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```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19044.1645]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Miles pc>ng new angular-fundamentals
? Would you like to add Angular routing? Yes
? Which stylesheet format would you like to use? CSS
CREATE angular-fundamentals/angular.json (3123 bytes)
CREATE angular-fundamentals/package.json (1083 bytes)
CREATE angular-fundamentals/README.md (1073 bytes)
CREATE angular-fundamentals/tsconfig.json (863 bytes)
CREATE angular-fundamentals/.editorconfig (274 bytes)
CREATE angular-fundamentals/.gitignore (548 bytes)
CREATE angular-fundamentals/.browserslistrc (600 bytes)
CREATE angular-fundamentals/karma.conf.js (1437 bytes)
CREATE angular-fundamentals/tsconfig.app.json (287 bytes)
CREATE angular-fundamentals/tsconfig.spec.json (333 bytes)
CREATE angular-fundamentals/.vscode/extensions.json (130 bytes)
CREATE angular-fundamentals/.vscode/launch.json (474 bytes)
CREATE angular-fundamentals/.vscode/tasks.json (938 bytes)
CREATE angular-fundamentals/src/favicon.ico (948 bytes)
CREATE angular-fundamentals/src/index.html (305 bytes)
CREATE angular-fundamentals/src/main.ts (372 bytes)
CREATE angular-fundamentals/src/polyfills.ts (2338 bytes)
CREATE angular-fundamentals/src/styles.css (80 bytes)
CREATE angular-fundamentals/src/test.ts (745 bytes)
CREATE angular-fundamentals/src/assets/.gitkeep (0 bytes)
```

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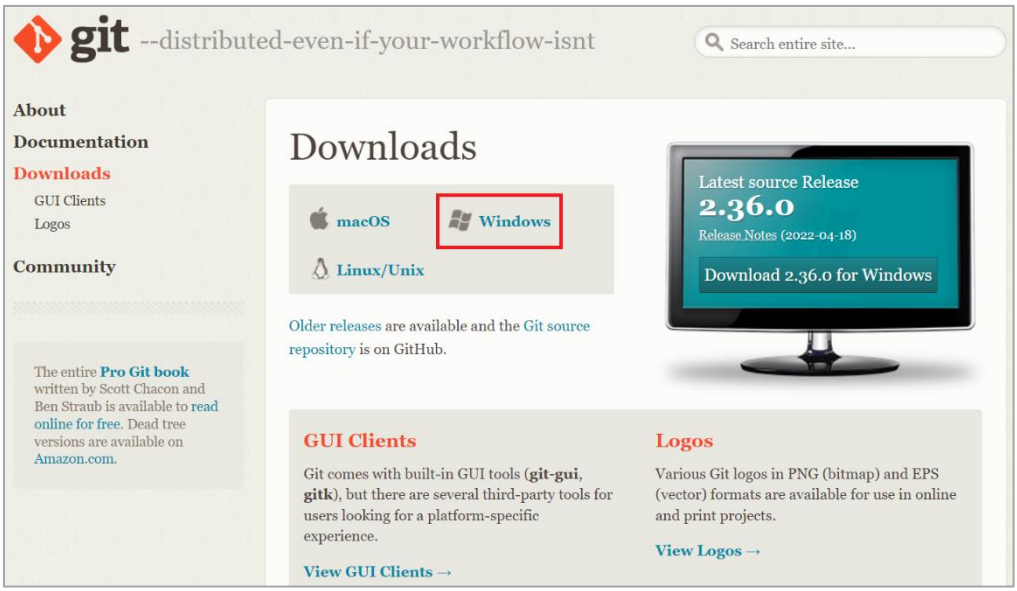
4. Git Installation

Git is a widely used open-source software tracking application used to track projects across different teams and revision levels.

Installing Git prompts you to select a text editor. If you don't have one, we strongly advise you to install it prior to installing Git.

- **Step 1: Download to the official website- https://git-scm.com/downloads**

Click the download link for Windows and allow the download to complete.



- **Step 2: Extract and Launch Git Installer**

Browse to the download location (or use the download shortcut in your browser).

Double-click the file to extract and launch the installer.

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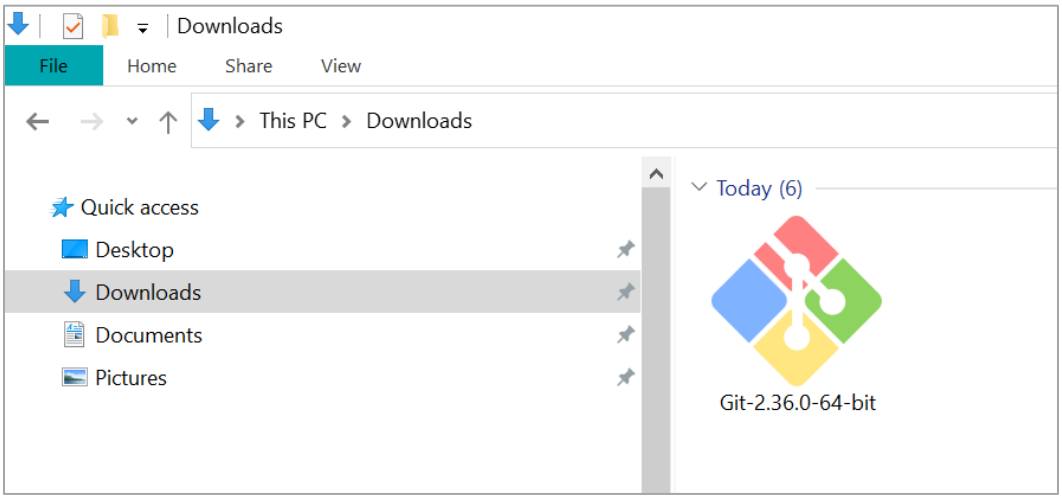
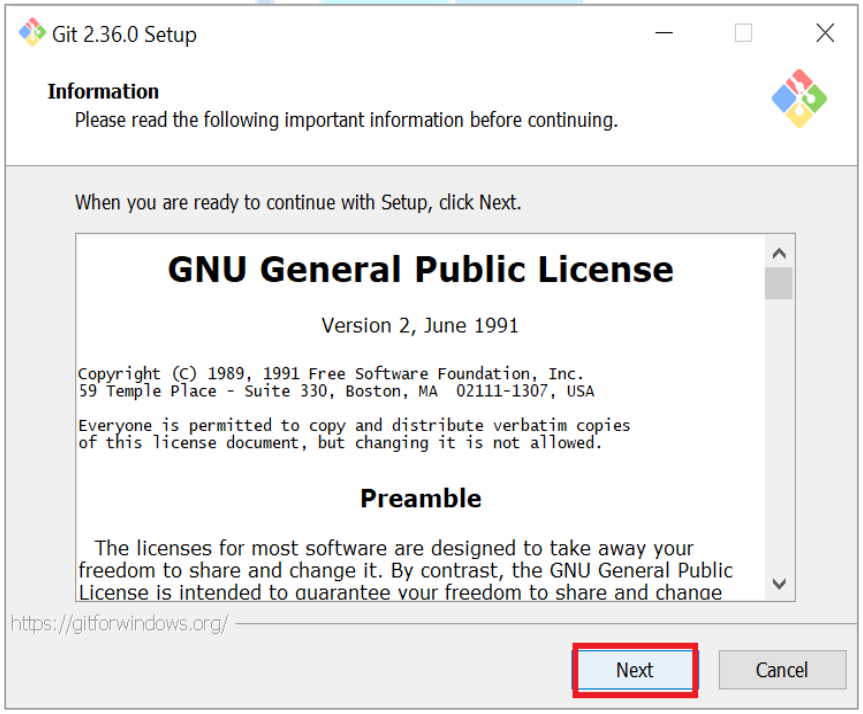


- **Step 3: Allow the app to make changes to your device by clicking Yes on the User Account Control dialog that opens.**
- **Step 4: Review the GNU General Public License, and when you're ready to install, click Next.**



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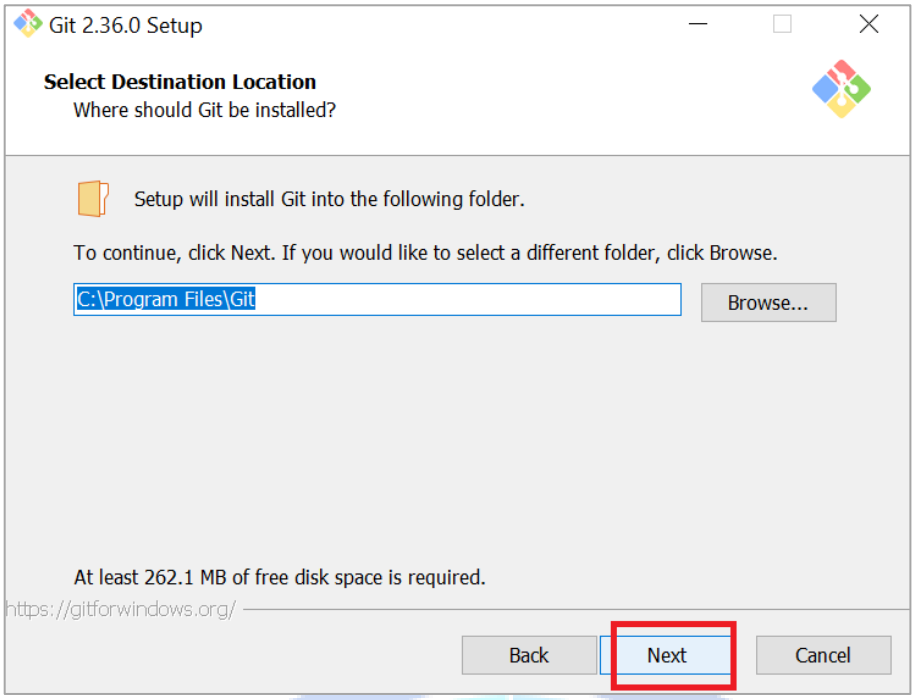
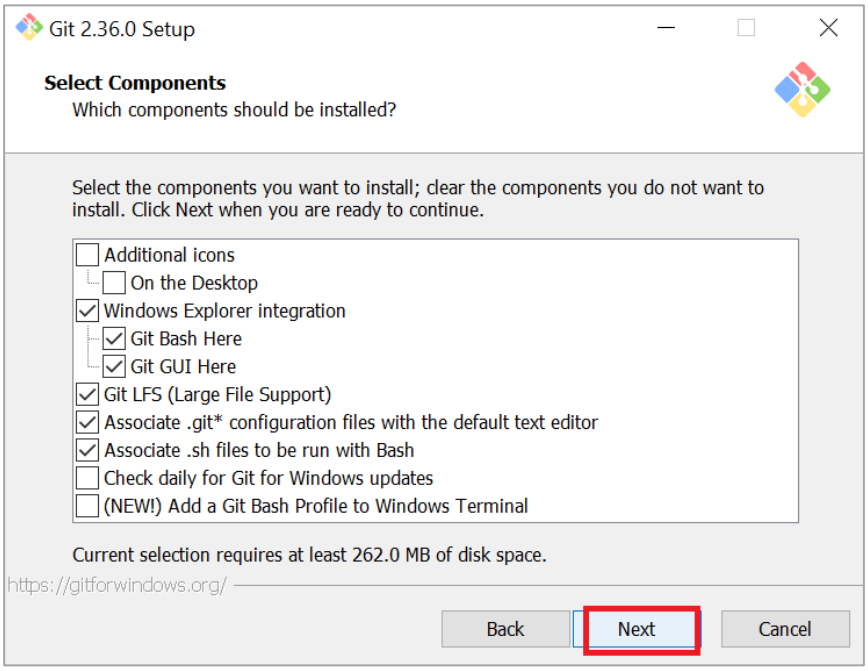
- **Step 5: The installer will ask you for an installation location. Leave the default, unless you have reason to change it, and click Next.**



- **Step 6: A component selection screen will appear. Leave the defaults unless you have a specific need to change them and click Next.**



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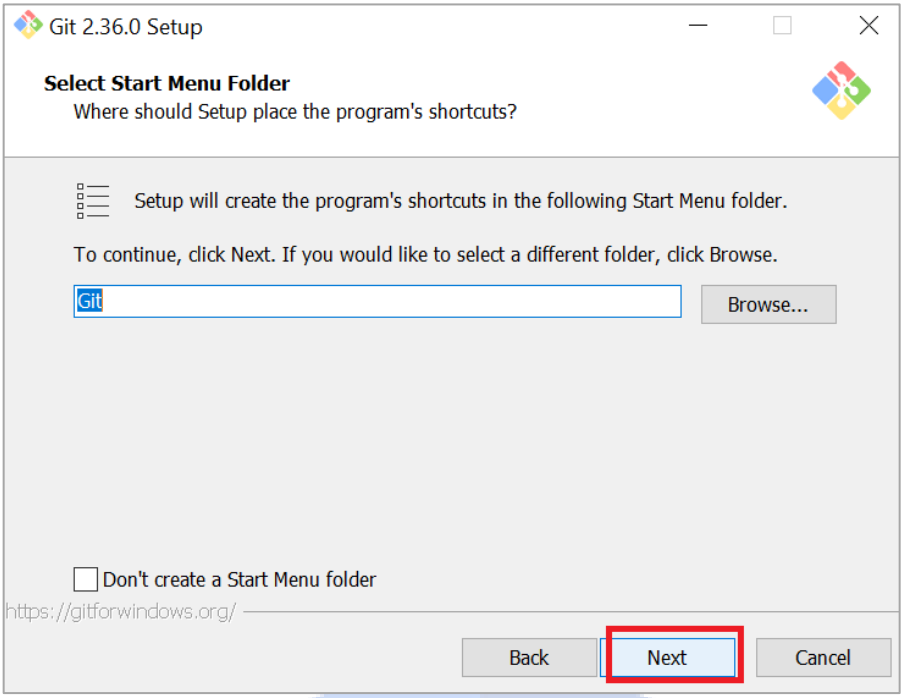
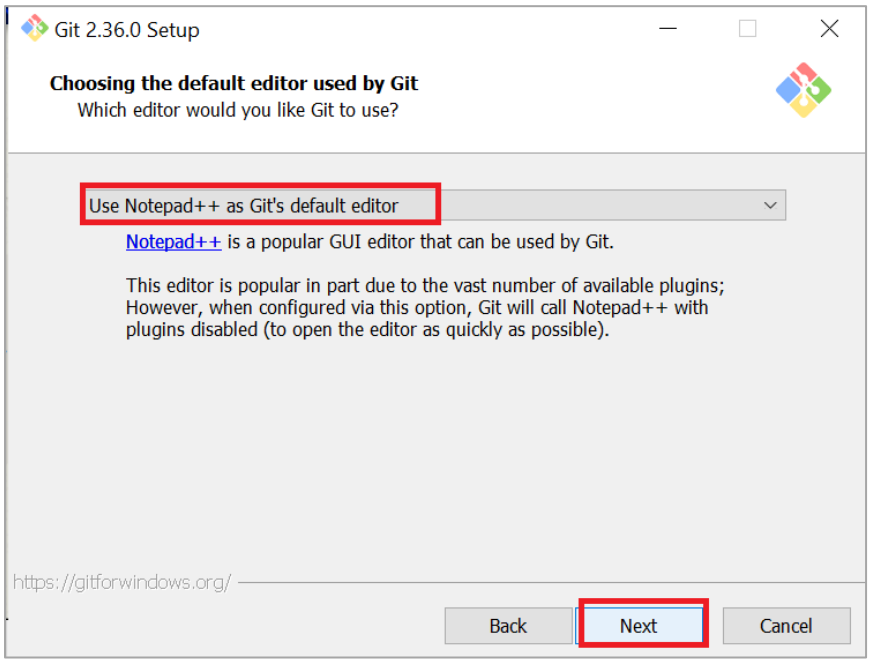
- **Step 7: The installer will offer to create a start menu folder. Simply click Next.**



- **Step 8: Select a text editor you'd like to use with Git. Use the drop-down menu to select Notepad++ (or whichever text editor you prefer) and click Next.**



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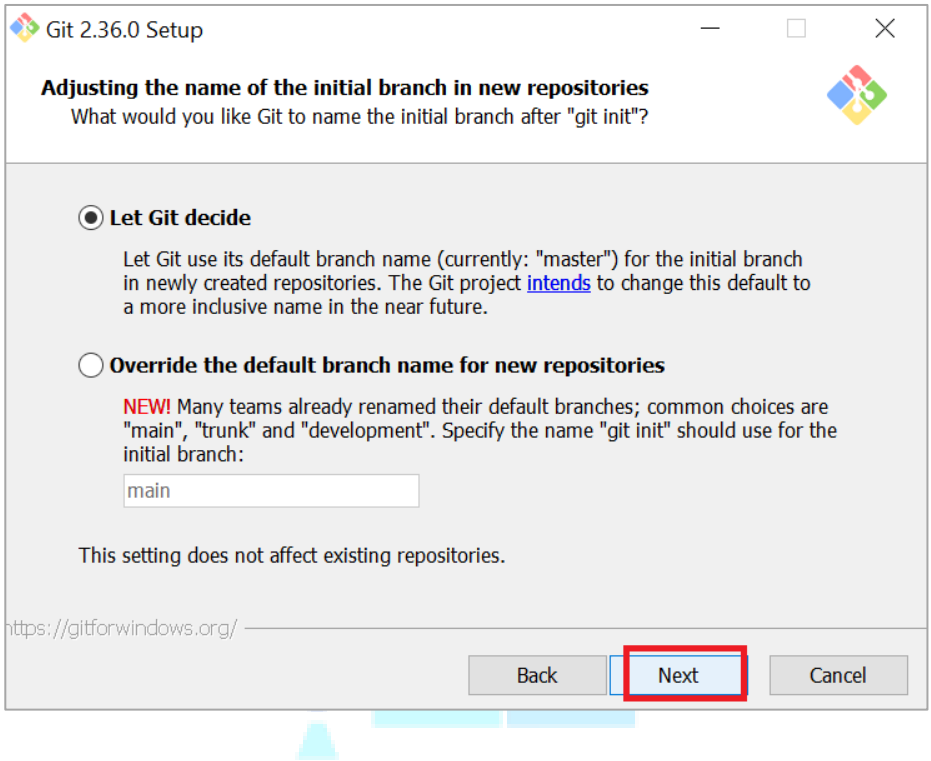
- **Step 9: The next step allows you to choose a different name for your initial branch. The default is 'master.' Unless you're working in a team that requires a different name, leave the default option and click Next.**



- **Step 10: This installation step allows you to change the PATH environment. The PATH is the default set of directories included when you run a command from the command line. Leave this on the middle (recommended) selection and click Next.**

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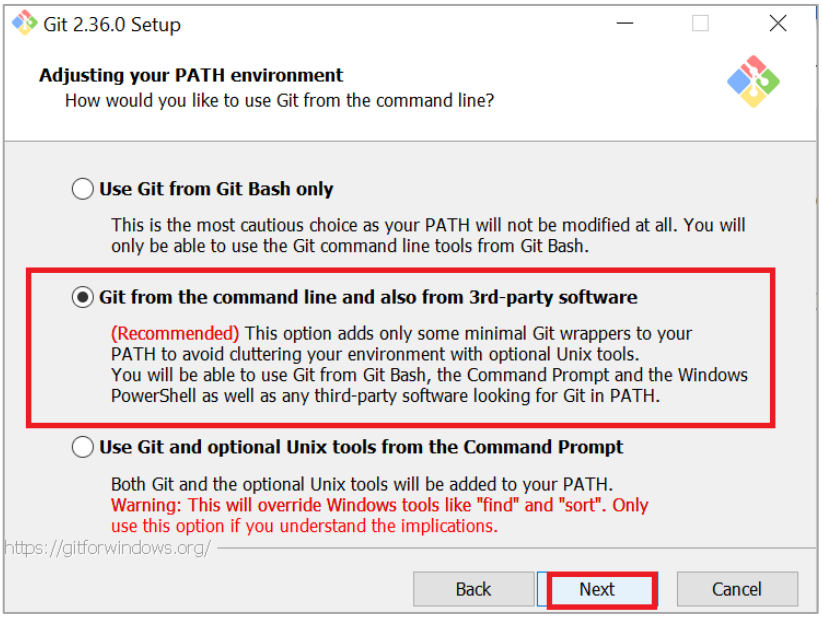
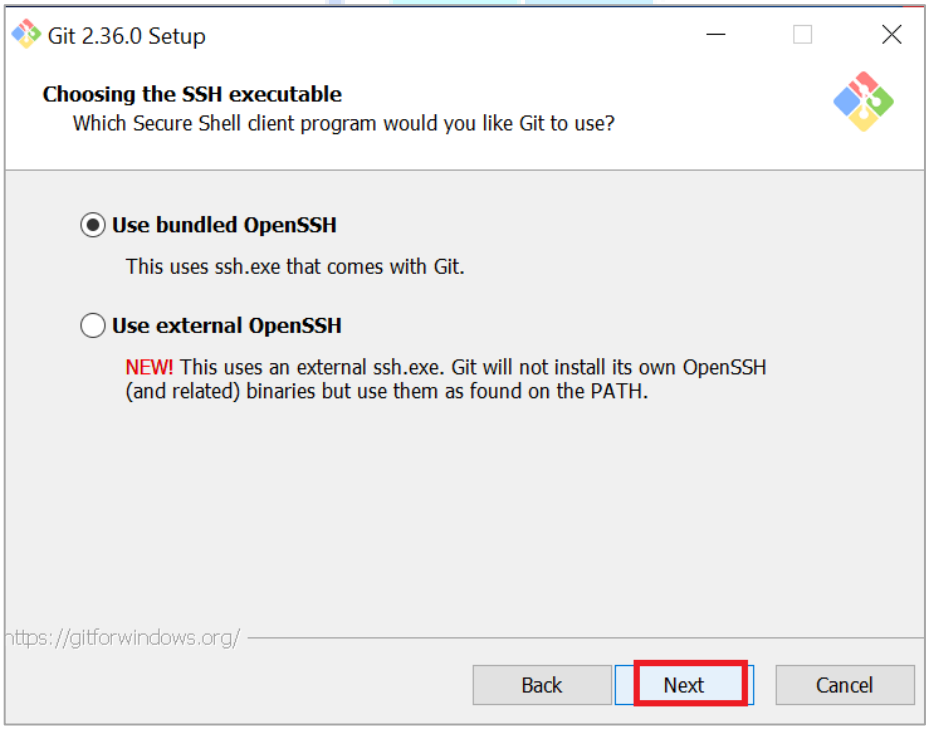


- **Step 11: Server Certificates, Line Endings and Terminal Emulators**

The installer now asks which **SSH client** you want Git to use. Git already comes with its own SSH client, so if you don't need a specific one, leave the default option and click **Next**.



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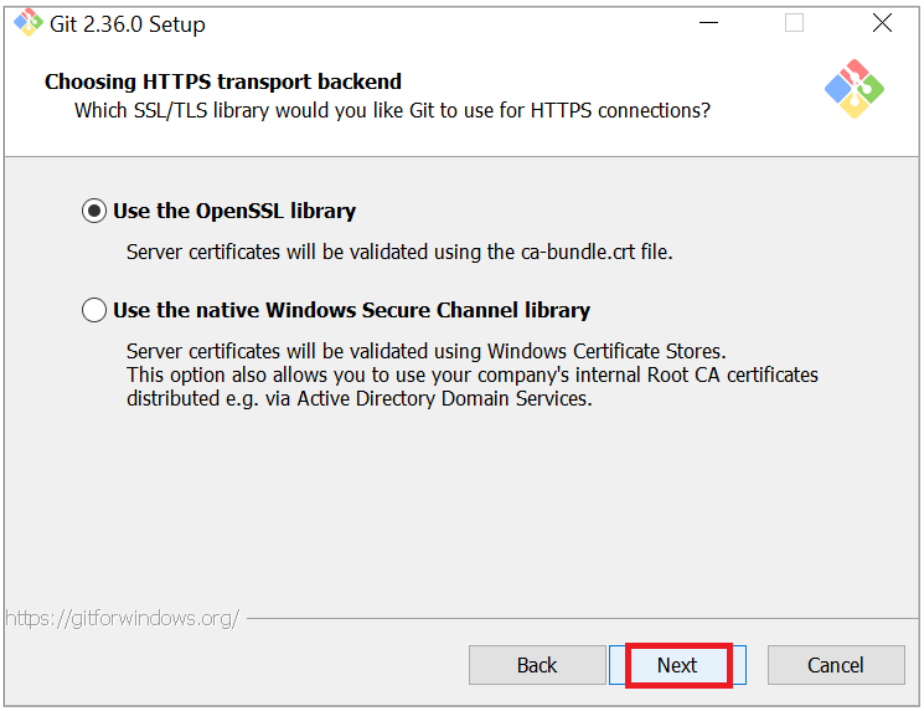
- **Step 12: This option is related to server certificates. Most users should use the default.**

If you're working in an Active Directory Environment, you may need to switch to Windows Store Certificates. Click Next.



- **Step 13: The next selection converts line endings. It is recommended that you leave the default selection.**

If you're working in an Active Directory Environment, you may need to switch to Windows Store Certificates. Click Next.

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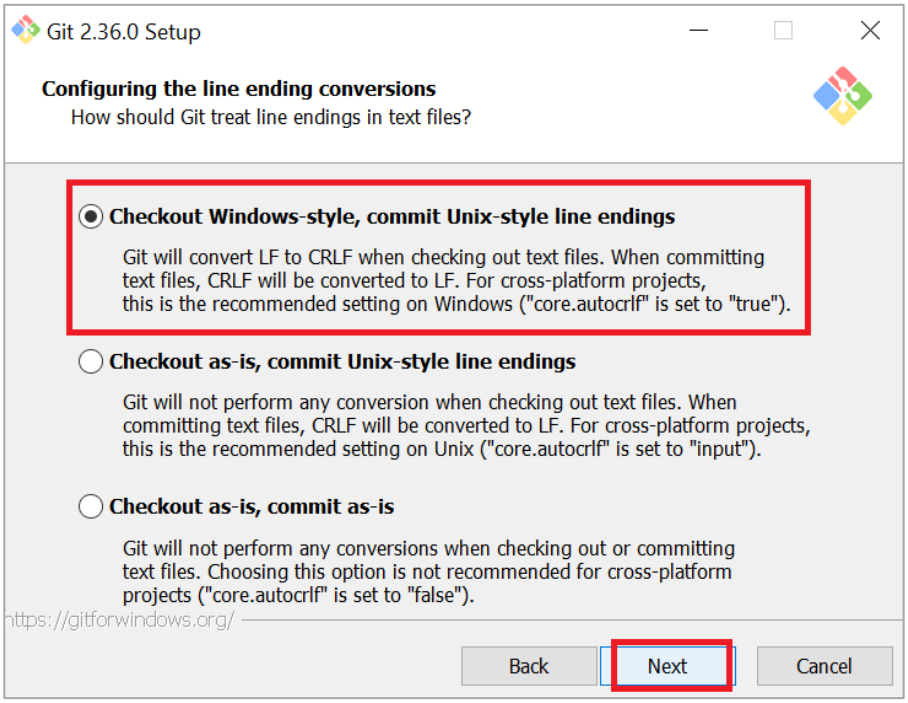
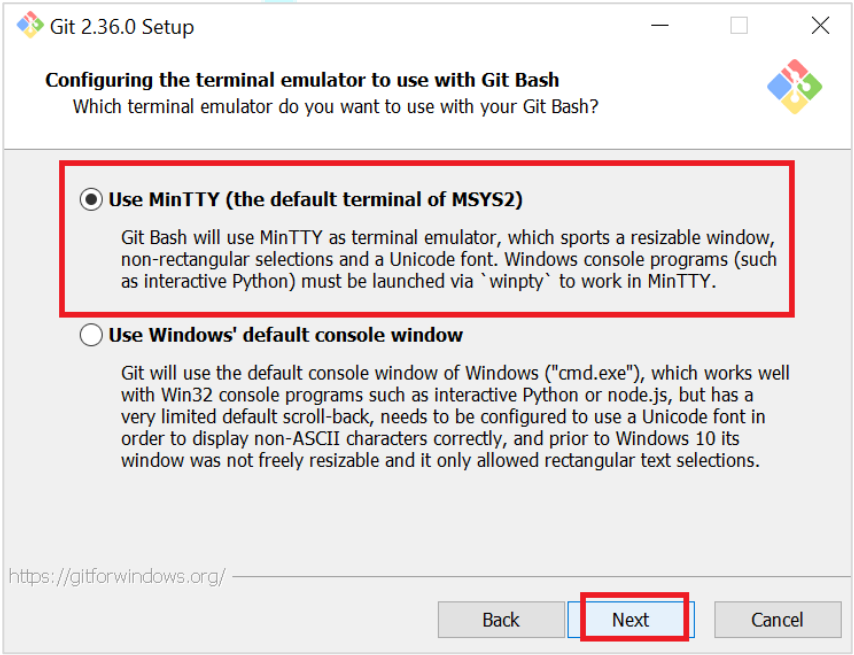


- **Step 14: Choose the terminal emulator you want to use. The default MinTTY is recommended, for its features. Click Next.**



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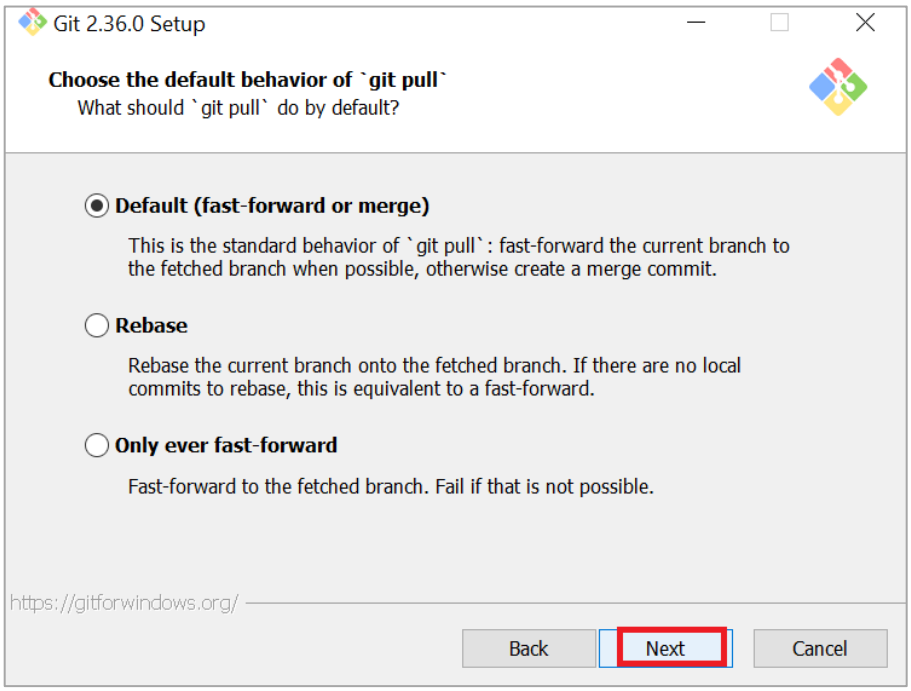
- **Step 15: The installer now asks what the git pull command should do. The default option is recommended unless you specifically need to change its behavior.**

Click Next to continue with the installation.



- **Step 16: Next you should choose which credential helper to use. Git uses credential helpers to fetch or save credentials.**

Leave the default option as it is the most stable one, and click Next.

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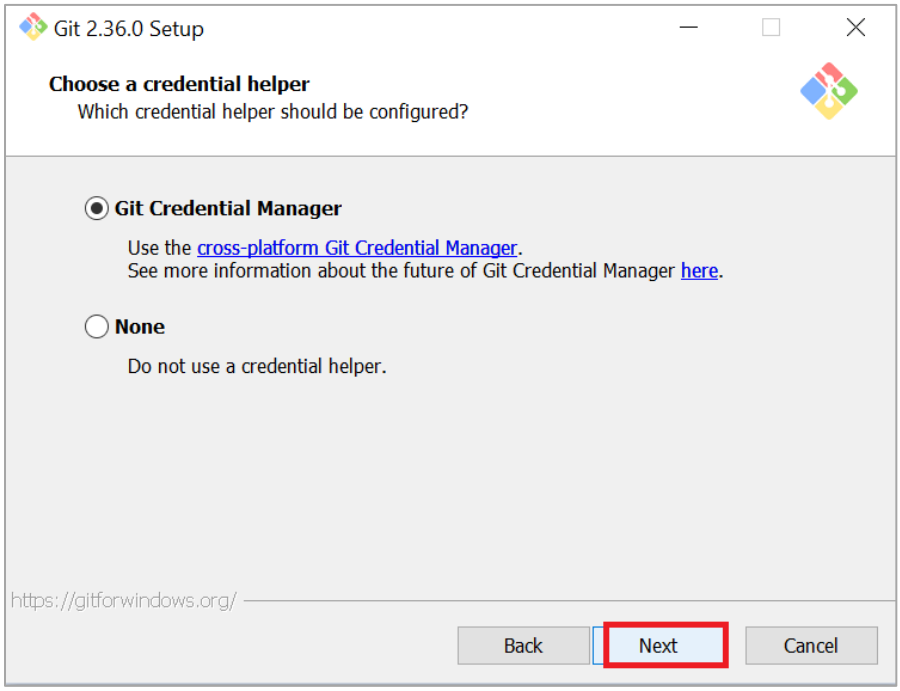

- **Step 17: Additional Customization Options**

The default options are recommended;

However this step allows you to decide which extra option you would like to enable. If you use symbolic links, which are like shortcuts for the command line, tick the box. Click **Next**.

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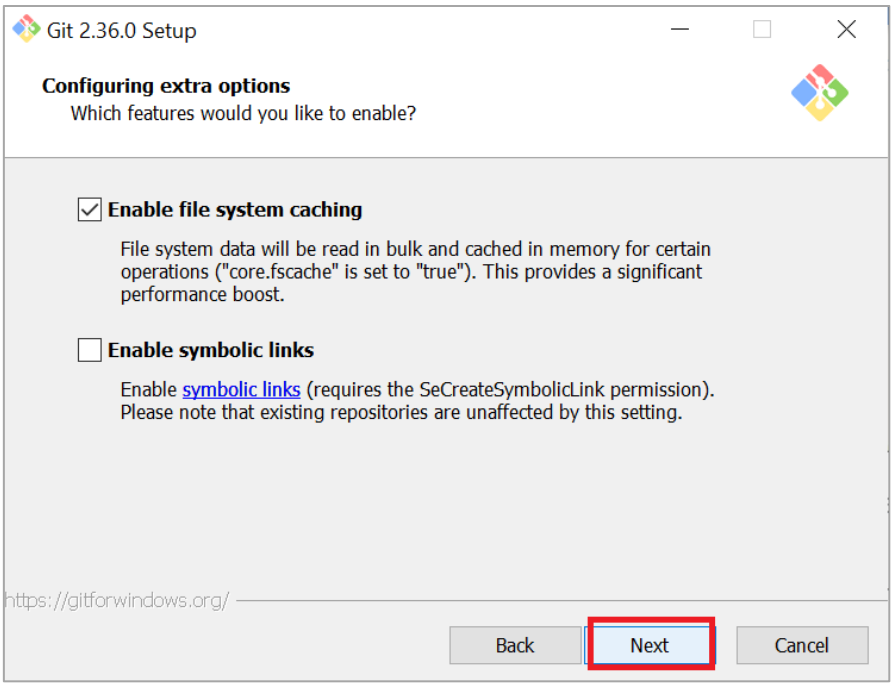

- **Step 18: Depending on the version of Git you're installing; it may offer to install experimental features.**

At the time this article was written, the options to include support for pseudo controls and a built-in file system monitor were offered.

Unless you are feeling adventurous, leave them unchecked and click Install.

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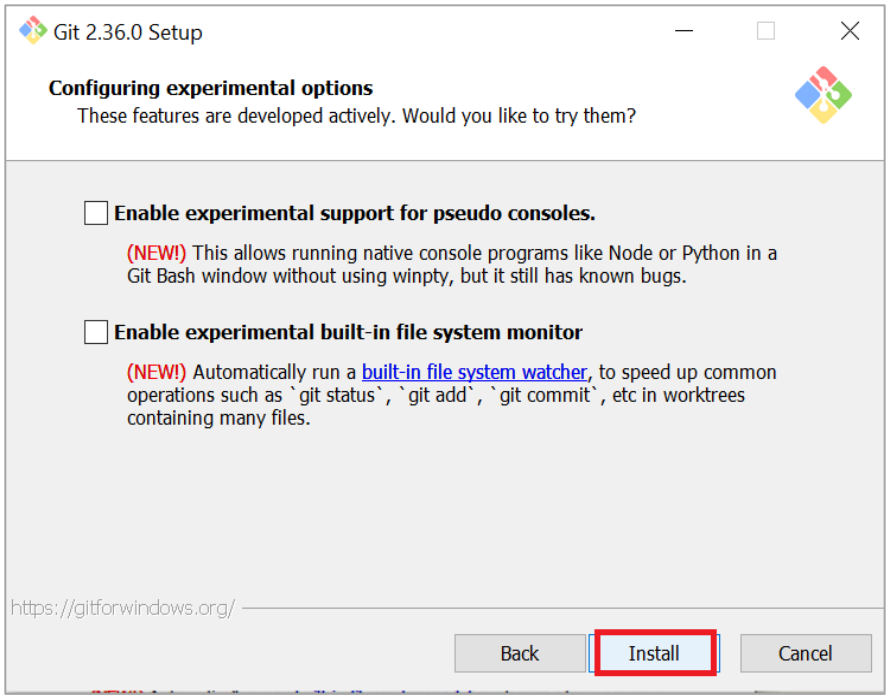

- **Step 19: Complete Git Installation Process**

Once the installation is complete, tick the boxes to view the Release Notes or Launch Git Bash, then click **Finish**.

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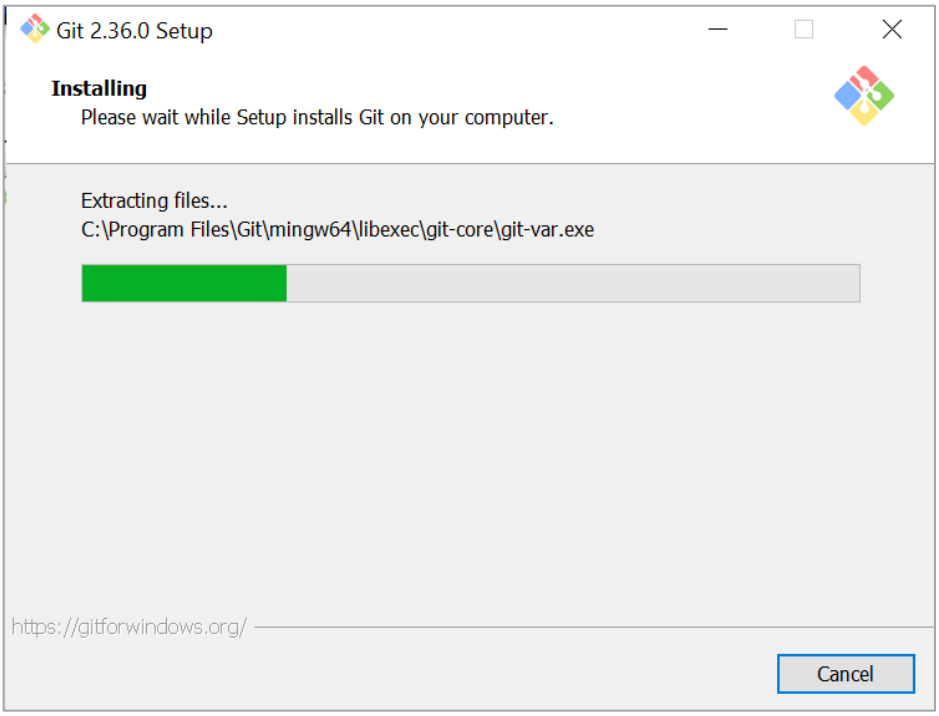

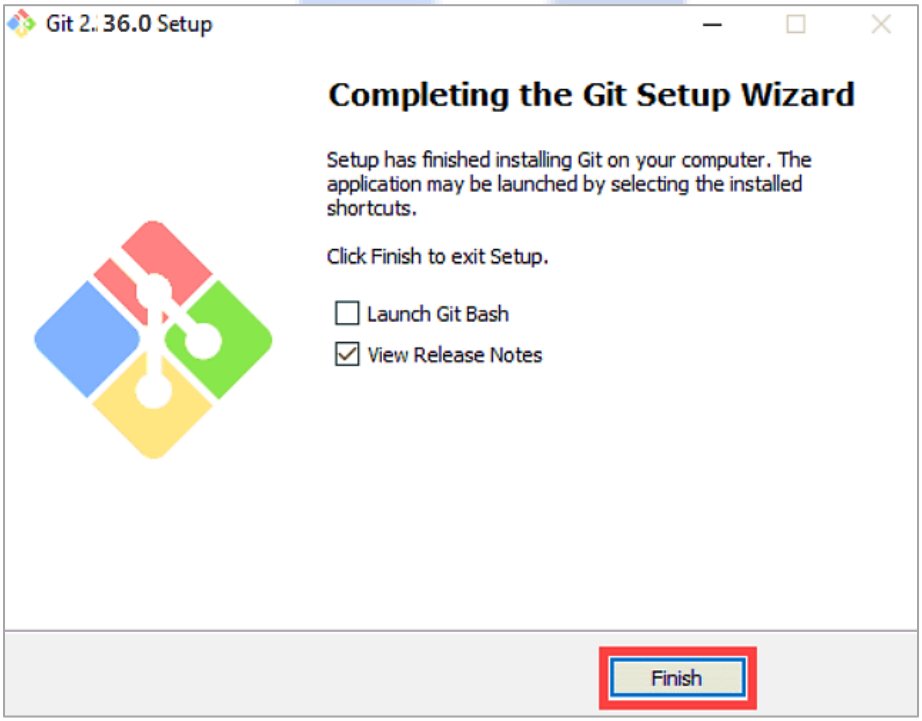

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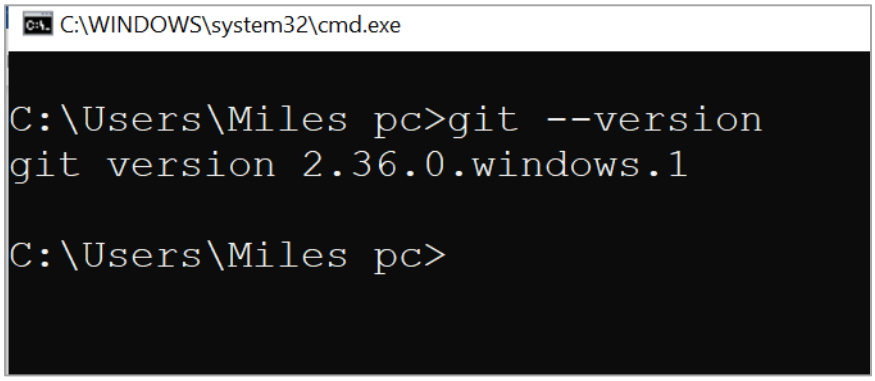
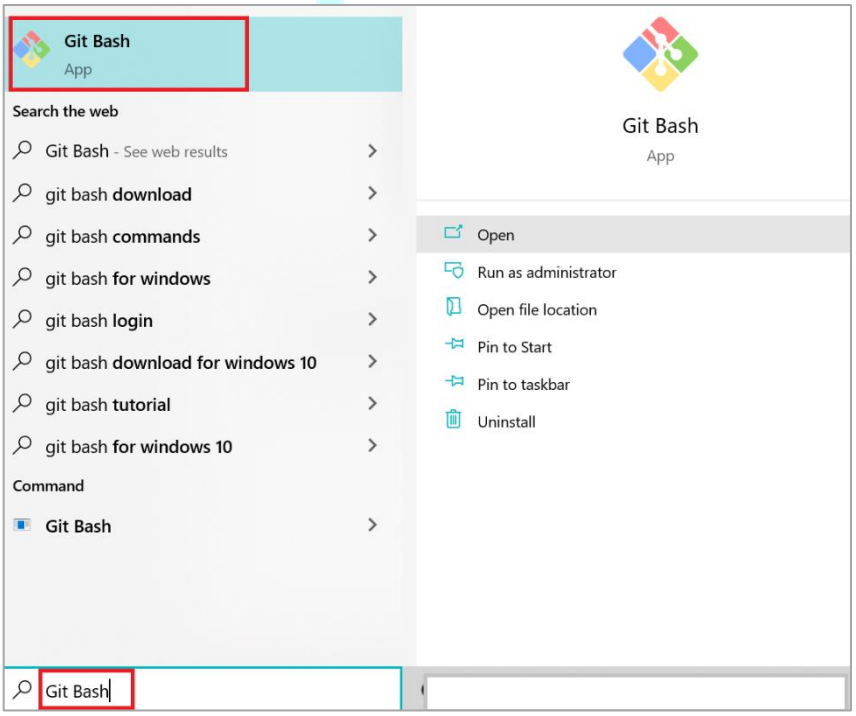
- **Step 20: Verify the Git version**

To check if Git has successfully installed on your system, run **git –version** command on the command prompt.



- **Step 21: Launch Git Bash Shell**

To launch **Git Bash** open the **Windows Start** menu, type *git bash* , and press **Enter** (or click the application icon).



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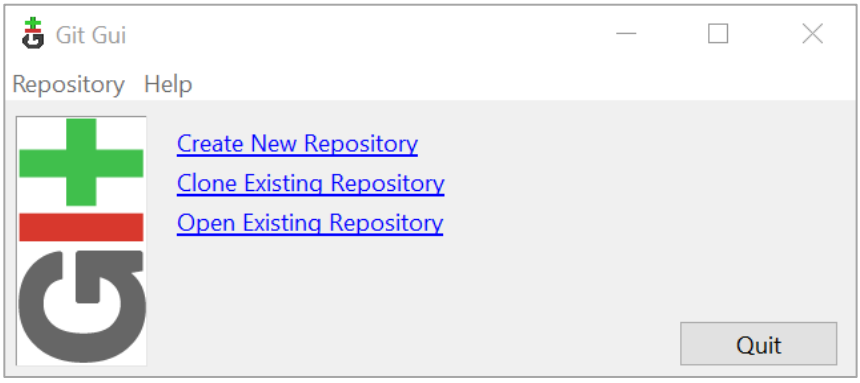
- **Step 22: Launch Git GUI**

To launch **Git GUI** open the **Windows Start** menu, type **git gui** and press **Enter** (or click the application icon).



You're all set!

Contact Us

If you are still facing any issues while installing the software and configuring the system, please free to reach out to us at **+91 9999 123 502/503** or email us at hello@scholarhat.com

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